

TUNGANA VINOD KUMAR

Visakhapatnam, Andhra Pradesh, India | tunganavinodkumar@gmail.com | +91 9346229360

LinkedIn | GitHub | Portfolio

Summary

Python Backend & GenAI Developer with a strong foundation in Computer Science (AI & ML). Experienced in designing REST APIs with FastAPI, Django, and Flask, building Retrieval-Augmented Generation (RAG) pipelines with LLMs, and managing relational SQL databases. Skilled in document processing and machine learning workflows, with a focus on writing clean, maintainable code for real-world applications.

Technical Skills

- **Programming Languages:** Python, SQL, Java
- **Frameworks & Libraries:** FastAPI, REST APIs, Django, Flask, Streamlit, NumPy, Pandas
- **AI & Generative AI:** Retrieval-Augmented Generation (RAG), LLMs (Llama 3.1, Gemini), OCR Document Processing (Tesseract, OCRmyPDF)
- **Web & Databases:** HTML, CSS, MySQL, SQLite
- **Core & Tools:** Object-Oriented Programming, Git, GitHub, VS Code, Google Colab

Work Experience

Andhra Pradesh State Skill Development Corporation (APSSDC) | Web Development Intern (Django)

Remote | Apr 2025 – Jun 2025

- Built database-backed web modules and REST endpoints using Python, Django, and SQLite with user authentication and role-based access.
- Managed application data, user accounts, and media files using Django's built-in admin panel and ORM queries.

EduSkills Foundation | AI & Machine Learning Virtual Intern

Remote | Jul 2025 – Sep 2025

- Preprocessed datasets, handled missing values, and built classification models using Python, Scikit-learn, Pandas, and NumPy.
- Evaluated model performance using accuracy, precision, and recall metrics to ensure reliable predictions.

Academic Projects

AI Examination Cell Chatbot & Document Intelligence System | FastAPI, Python, RAG, SQLite, React

- Built an AI chatbot using FastAPI and Groq Llama 3.1 to answer student examination inquiries from official college notices.
- Created an end-to-end RAG pipeline using Tesseract OCR and vector retrieval to extract text from PDF circulars and generate grounded answers.

SoundWave — Python & Django Audio Streaming Application | Python, Django, SQLite, HTML, CSS

- Developed a music streaming web app using Python, Django, and SQLite featuring playlist management and audio playback controls.
- Designed relational database tables for users, tracks, and playlists with Django ORM for fast queries.

Liver Disease Clinical Risk Predictor | Python, Flask, Scikit-learn, Pandas

- Developed a machine learning web app using Flask and Scikit-learn to predict liver disorder risks from patient clinical test values.
- Handled data cleaning and feature scaling using Pandas and NumPy, deploying the model with a web interface on PythonAnywhere.

Education

Nadimpalli Satyanarayana Raju Institute of Technology (NSRIT Autonomous) | Oct 2022 – May 2026

B.Tech in Computer Science & Engineering (AI & ML) | CGPA: 8.33 / 10

Sri Sampath Sai Junior College | Jul 2020 – Apr 2022

Intermediate (Mathematics, Physics & Chemistry) | Percentage: 77.5%

Certifications

- Google Cloud AI Foundations – LLMs & Generative AI (Google Cloud)
- 5-Day AI Agents Intensive Course (Kaggle × Google)
- Data Analysis with Python (IBM / Coursera)
- Machine Learning with Python (IBM SkillsBuild)